CalCGPA — Calculation Guidance

Input entry explanations & concise formulas for all sections • Developed by **R.Nithin Annamalai**

1. Semester GPA (Grade Point Average)

Input Entries:
- **Course Name:** Title / code of the subject.
- **Credits:** Subject credit weight (e.g. 1, 2, 3, 4).
- **Grade:** Letter grade received (O to U).

Formula:

GPA = Σ (Credits$_i$ × GradePoints$_i$) / Σ Credits$_i$

Total Points = sum of (Credits × Points for each course)

Total Credits = sum of all course credits

Grade	Mark Range	Points	Rating
O	91 – 100	**10**	Outstanding
A+	81 – 90	**9**	Excellent
A	71 – 80	**8**	Very Good
B+	61 – 70	**7**	Good
B	56 – 60	**6**	Above Average
C	50 – 55	**5**	Average
U	< 50	**0**	Re-appear / Fail

2. Overall CGPA (Cumulative Grade Point Average)

Mode A: GPA Only (Simple Average)

Input Entries:
- **Semester GPAs:** Enter GPA values separated by comma or newline (e.g. 8.5, 9.2, 7.8).

Formula:

CGPA = (Σ GPA$_i$) / N

Where N = number of semesters entered.

Mode B: Credit-Weighted CGPA (Official)

Input Entries:
- **GPA & Credits:** Enter semester GPA and total credits registered for each semester row.

Formula:

CGPA = Σ (GPA$_i$ × Credits$_i$) / Σ Credits$_i$

Weights semesters by their respective credit counts.

3. Required GPA (Target CGPA Planner)

Mode 1: Semester-Count Estimation

Input Entries:
- **Current CGPA:** Cumulative GPA earned so far.
- **Target CGPA:** Desired cumulative CGPA goal.
- **Semesters Completed (N):** Number of terms finished.

Formula:

Required GPA = Target × (N + 1) – Current × N

Mode 2: Credit-Weighted (Accurate)

Input Entries:
- **Current CGPA:** Existing cumulative GPA.
- **Target CGPA:** Desired overall CGPA goal.
- **Completed Credits (C_{prev}):** Total credits earned so far.
- **Upcoming Credits (C_{next}):** Credits in the upcoming term.

Formula:

Required GPA = [Target × (C_{prev} + C_{next}) – Current × C_{prev}] / C_{next}

4. End Semester Marks & Grading

Assessment Input Entries & Total Formula

Input Entries:
- **Split Mode (40:60):** Internal Test 1 (/50), Internal Test 2 (/50), Term Work (/10), End Sem Exam (/100).
- **Total Mode:** Total Internals (/40 or /50), End Sem Exam (/100).

Total Marks (/100) Formula:
- **Split Mode:** (Test 1 × 0.3) + (Test 2 × 0.3) + Term Work + (End Sem × 0.6)
- **Total /40 Mode:** Internals (/40) + (End Sem × 0.6)
- **Total /50 Mode:** Internals (/50) + (End Sem × 0.5)

Target End Sem Goal Formula

Input Entry:
- **Want a total of?:** Desired final score out of 100.

Required End Sem Exam Score (/100) Formula:

Needed End Sem = (Target Total – Internals) / Factor

- **Factor = 0.6** for 40:60 assessment mode
- **Factor = 0.5** for 50:50 assessment mode